



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Department of Electrical, Electronic and
Computer Engineering

Semester Test 2-Memo

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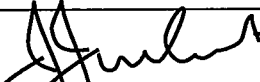
Module EBN111

Electricity and Electronics

24 April 2009

Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	45	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 minute <i>90 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Dosent(e): <i>Lecturer(s):</i>	1 J Joubert		
	2 DV Bhatt		
Groephoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding. <i>Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.</i>			
Groephoof: <i>Group Head:</i>	Prof J Joubert	Handtekening: <i>Signature:</i>	

VRAAG/QUESTION 1 [15]

Hierdie vraag moet op die OPTIESE MERKBLAD (kant 2) geantwoord word. Kies die mees korrekte antwoord.	This question must be answered on the OPTICAL MARK SHEET (side 2). Choose the answer which is most correct.
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Vraag 1.1 [1] The ooplus-versterking van 'n OV is gedefinieer as die: A. Verhouding van uitsetspanning tot uitsetweerstand B. Verhouding van uitsetweerstand tot insetweerstand C. Verhouding van uitsetspanning tot verskil van insetspanning D. Verhouding van insetspanning tot verskil van uitsetspanning E. Verhouding van verskil van insetspanning tot uitsetspanning	Question 1.1 [1] The open-loop gain of an op amp is defined as the: A. Ratio of output voltage to output resistance B. Ratio of output resistance to input resistance C. <u>Ratio of output voltage to difference of input voltage</u> D. Ratio of input voltage to difference of output voltage E. Ratio of difference of input voltage to output voltage
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Vraag 1.2 [1] Watter stelling oor 'n nie-ideale OV is waar? A. Die ooplus-versterking is ∞ B. Die insetweerstand is 0 C. Die uitsetweerstand is hoog D. Die uitsetspanning is gewoonlik baie hoog E. Die uitsetweerstand is gewoonlik laag F. Die insetspanning moet hoog wees	Question 1.2 [1] Which statement about a non-ideal op amp is true? A. The open-loop gain is ∞ B. Its input resistance is 0 C. Its output resistance is high D. The output voltage is usually very high E. <u>The output resistance is usually low</u> F. The input voltage should be high
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Vraag 1.3 [1] Watter stelling oor 'n OV wat in die lineêre gebied werk is waar? (V_{cc} is die toevoerspanning) A. $ \text{verskil van die insetspannings} \geq \text{uitsetspanning}$ B. $ \text{verskil van die insetspannings} < V_{cc}$ C. $ \text{uitsetspanning} \geq V_{cc}$ D. $ \text{uitsetspanning} \leq V_{cc}$ E. Geen van bogenoemde nie	Question 1.3 [1] <i>statement</i> Which/about an op amp operating in linear range is true? (V_{cc} is the source voltage) A. $ \text{input differential voltage} \geq \text{output voltage}$ B. $ \text{input differential voltage} < V_{cc}$ C. $ \text{output voltage} \geq V_{cc}$ D. <u>$\text{output voltage} \leq V_{cc}$</u> E. None of the above.
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Vraag 1.4 [1] Watter stelling oor 'n omkeer OV-baan is nie waar nie? A. Die spanningsversterking neem toe as die terugvoerweerstand groter word B. Die spanningsversterking neem af as die intree weerstand groter word C. Die spanningsversterking neem toe as die verhouding van die terugvoer tot intree weerstand groter word D. Die spanningsversterking neem toe as die verhouding van die terugvoer tot intree weerstand kleiner word E. Die spanningsversterking neem af as die verhouding van die intree tot terugvoerweerstand groter word	Question 1.4 [1] Which statement about an inverting op amp circuit is not true? A. The voltage gain increases with the increase in feedback resistance B. The voltage gain decreases with the increase in input resistance C. The voltage gain increases when the ratio of feedback resistance to input resistance is increased D. <u>The voltage gain increases when the ratio of feedback resistance to input resistance is decreased</u> E. The voltage gain decreases when the ratio of input resistance to feedback resistance is increased
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Question 1.5 [1]

Calculate V_o in Fig Q1.5, if $R_1=0$ and $R_2=\infty$:

- A. 2V B. -2V C. 0 D. 6V E. None of the above

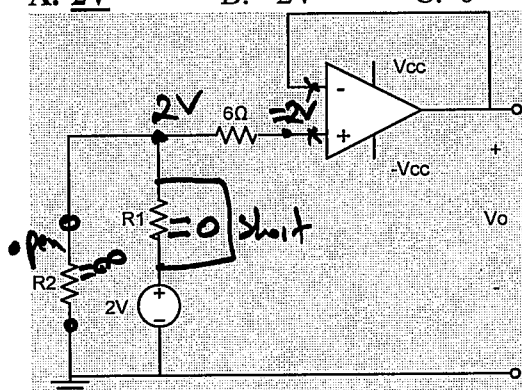


Fig. Q1.5

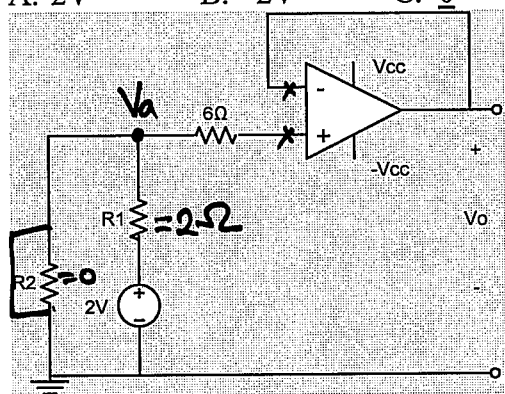
$R_1=0 \Rightarrow$ short circuit
 $R_2=\infty \Rightarrow$ open circuit.

$\therefore$ No voltage drop across 6Ω
 $\therefore V_o = 2V$ voltage follower

Question 1.6 [1]

Calculate V_o in Fig Q1.6, if $R_1=2\Omega$ and $R_2=0$:

- A. 2V B. -2V C. 0 D. 6V E. None of the above.



Q1.6

$R_2=0 \Rightarrow$ short circuit

$\therefore V_a = \text{Ground} = 0$

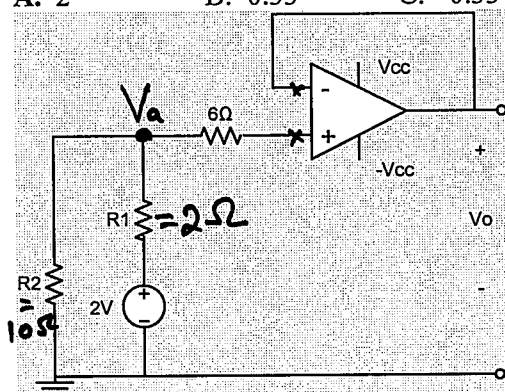
$\therefore$ No drop across 6Ω

$\therefore V_o = 0V$ (voltage follower)

Question 1.7 [1]

Calculate the voltage gain in Fig Q1.7 when $R_1=2\Omega$ and $R_2=10\Omega$.

- A. 2 B. 0.33 C. -0.33 D. 1.167 E. None of the above.



Q1.7

V_a is the voltage across R_2
 using voltage divider

$$V_a = \frac{10}{10+2} \cdot 2 = \frac{10}{6} = 1.667V$$

No drop across 6Ω .

$\therefore V_o = 1.667V$

Question 1.8 [1]

What value of R_2 can change the circuit of Fig. Q1.8 to a simple inverter for the 20mV source?

- A. Any value of R_2 that is not same as R_1
 E.. $R_2 = R_f$ F. None of the above.

B. When $R_1 = R_2$

C. $0 \leq R_2 \leq R_1$

D. $R_2 = \infty$

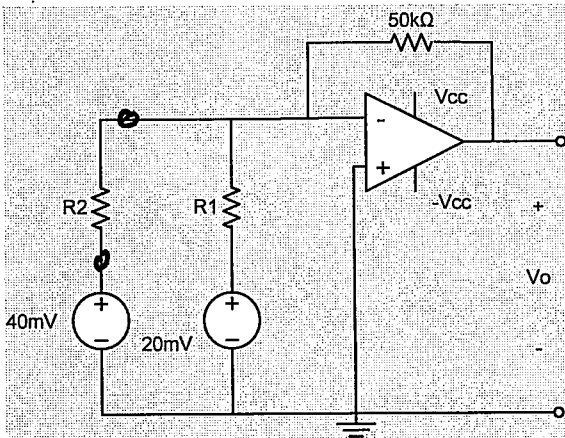


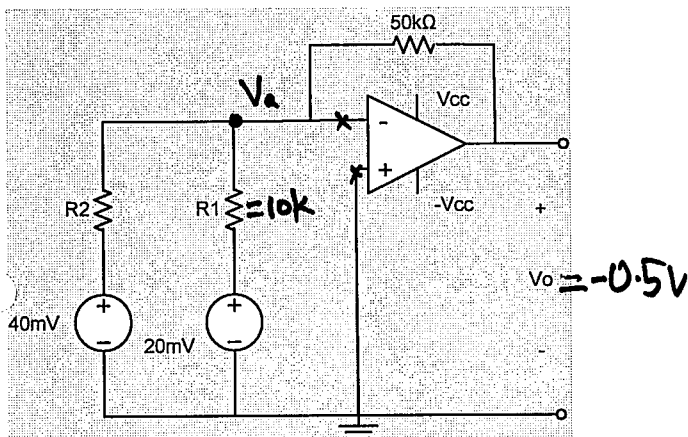
Fig. Q1.8, Fig. Q1.9

Circuit becomes simple inverter if R_2 has no influence
 $\therefore \underline{R_2 = \infty}$ (Open circuit.)

Question 1.9 [1]

Determine R_2 in Fig. Q1.9 if $V_0 = -0.5V$, and $R_1 = 10k\Omega$.

- A. $10k\Omega$ B. $5k\Omega$ C. 100Ω D. 500Ω E. None of the above.



Summer.

$$V_0 = - \left[\frac{50k}{10k} \cdot 20m + \frac{50k}{R_2} \cdot 40m \right]$$

$$= +100m + \frac{2000}{R_2} = -0.5$$

$$100m R_2 + 2000 = 0.5 R_2$$

$$\therefore (0.5 - 100m) R_2 = 2000$$

$$\therefore R_2 = \frac{2000}{0.4}$$

$$R_2 = 5000$$

$$= \underline{5k\Omega}$$

Question 1.10 [1]

Find the maximum power that can be delivered by the circuit in Fig Q1.10.

- A. 0.125 W B. 0.18 W C. 0.333 W D. 0.75 W E. 1.08 W F. None of the above

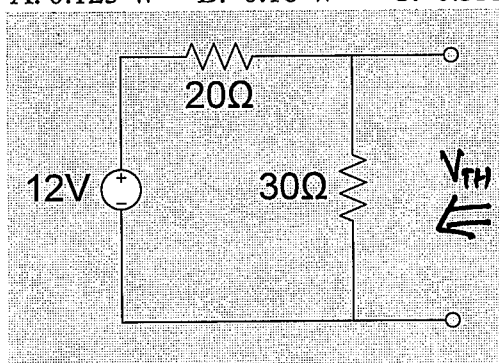
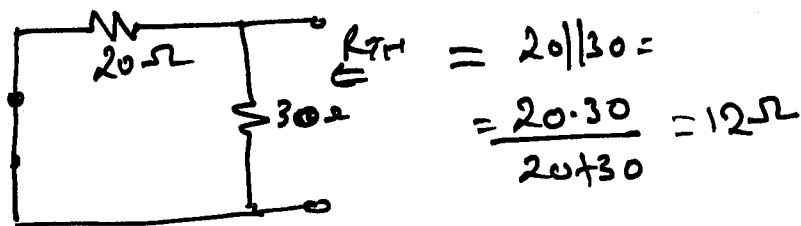


Fig Q1.10



$$V_{TH} = \frac{30}{20+30} \cdot 12$$

$$= 7.2V$$

$$\therefore P_{max} = \frac{V_{TH}^2}{4R_{TH}} = \frac{7.2^2}{4 \cdot 12}$$

$$\therefore P_{max} = 1.08W$$

Question 1.11 [1]

Find V_o for the circuit in Fig Q1.11?

- A. 5V B. -5V C. -10V D. 15V E. -15V F. None of the above.

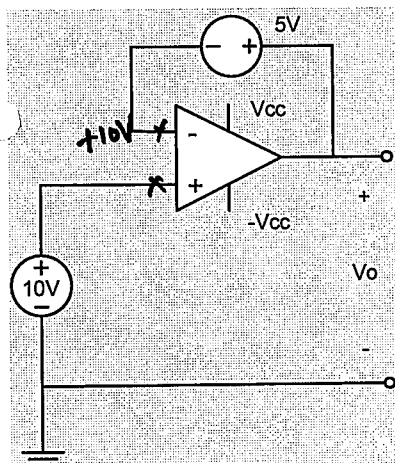


Fig. Q1.11

KVL:

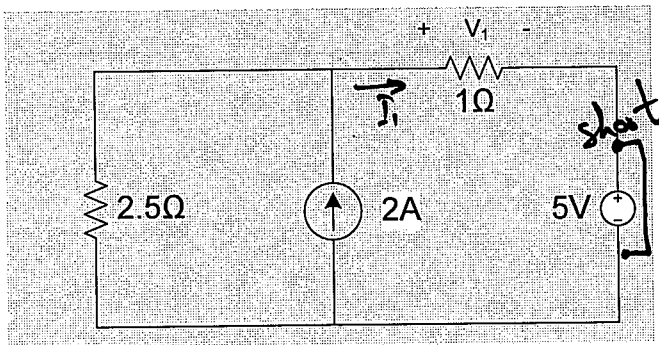
$$-10 - 5 + V_o = 0$$

$$\therefore V_o = \underline{-15V}$$

Question 1.12 [1]

Determine the value of V_1 due to the current source alone in Fig Q1.12.

- A. 1.43 V B. 2.86 V C. 2 V D. -1.43 V E. -2.86 V F. 0 V G. None of the above



Using current divider

$$I_1 = \frac{2.5}{1+2.5} \cdot 2 = 1.43A$$

$$\therefore V_1 = 1 \cdot 1.43 = \underline{1.43V}$$

Fig. Q1.12,

Question 1.13 [1]

Determine the value of V_1 due to the voltage source alone in Fig Q1.13.

- A. 1.43 V B. 2.86 V C. 2 V D. -1.43 V E. -2.86 V F. 0 V G. None of the above

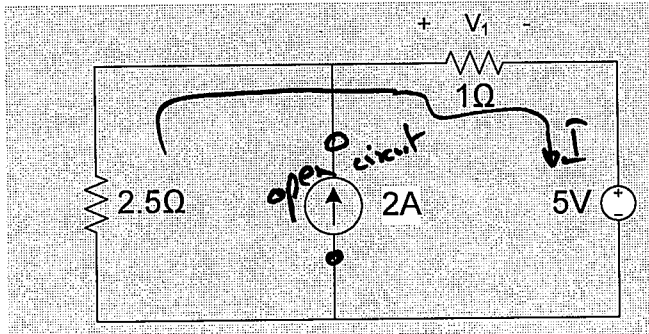


Fig Q1.13

$$2.5I + 1I + 5 = 0$$

$$3.5I + 5 = 0$$

$$I = \frac{-5}{3.5} = -1.43A$$

$$\therefore V_1 = I \cdot 1$$

$$= (-1.43) \cdot 1$$

$$V_1 = \underline{\underline{-1.43V}}$$

Question 1.14 [1]

Consider a bipolar junction transistor with $\alpha = 0.99$ in Fig Q1.14. The three currents are possibly:

- A. $I_1 = 1\text{mA}$, $I_2 = 99\text{mA}$, $I_3 = 1\text{A}$
- B. $I_1 = 1\text{mA}$, $I_2 = 999\text{mA}$, $I_3 = 1\text{A}$
- C. $I_1 = 10\text{mA}$, $I_2 = -990\text{mA}$, $I_3 = 1\text{A}$
- D. $I_1 = 10\text{mA}$, $I_2 = 990\text{mA}$, $I_3 = -1\text{A}$
- E. None of the above

$$I_2 = \beta I_1$$

$$I_E = (1 + \beta) I_B$$

$$I_E = I_B + I_C$$

$$I_C = \alpha I_E = \beta I_B$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

$$\alpha = \frac{\beta}{1 + \beta}$$

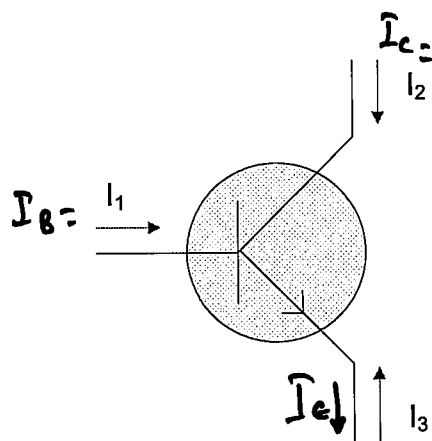


Fig Q1.14

Question 1.15 [1]

A Bipolar junction transistor is a

- A. dependent current controlled current source
- B. independent current controlled current source
- C. dependent voltage controlled current source
- D. dependent current controlled voltage source
- E. independent current controlled voltage source

Question 2.1 [4]

Given that $\beta=100$ and $V_{BE}=0.7V$, determine the following for the circuit in Fig.

Q2.1:

- a) I_B b) I_C c) V_O .

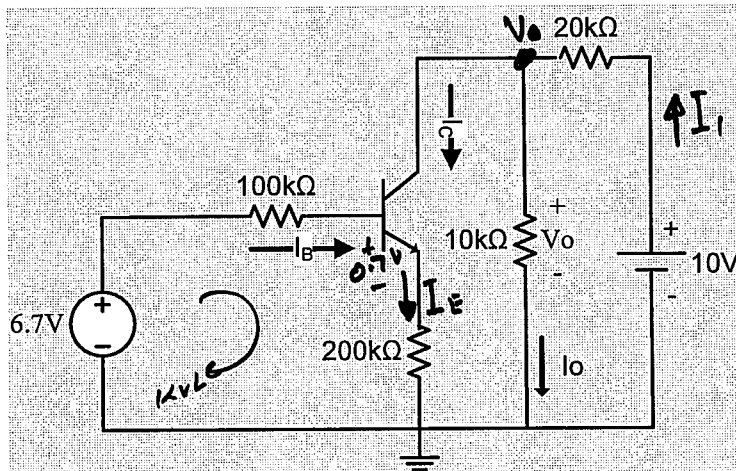


Fig. Q2.1

$$I_E = I_B + I_C$$

$$I_C = \alpha I_E \approx \beta I_B$$

$$I_E = (1 + \beta) I_B$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

$$\alpha = \frac{\beta}{1 + \beta}$$

a) $-6.7 + 100k I_B + 0.7 + 200k I_B = 0$

but $I_E = (1 + \beta) I_B = 101 I_B$

$-6.7 + 100k I_B + 200k (101) I_B = 0$

$\therefore I_B = \frac{6}{10k + 200k \cdot 101}$

$I_B = 0.296 \mu A = 296 nA$

b) $I_C = \beta I_B = 100 \cdot 0.296 \mu$

$I_C = 29.6 \mu A$

c) @ Node V_O : $I_1 = I_C + I_O$

$\frac{10 - V_O}{20k} = 29.6 \mu + \frac{V_O}{10k}$

$10 - V_O = 20k \cdot 29.6 \mu + 2V_O$

$\therefore 10 - 20k \cdot 29.6 \mu = 3V_O$

$\therefore V_O = \frac{10 - 20k \cdot 29.6 \mu}{3}$

$= 3.136 V$

Question 2.2 [3]

If $V_i = 10V$, determine the following for the circuit in Fig. Q2.2:

- a) V_a b) The gain $A_v = V_o/V_i$ c) I_o

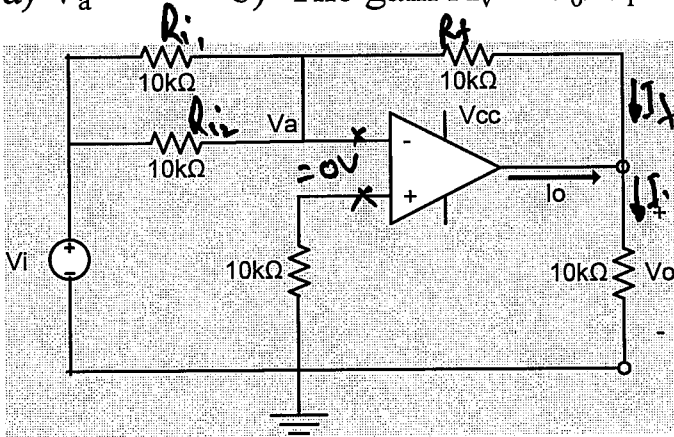


Fig. Q2.2

a) Since $10k\Omega$ is grounded
 $V_a = 0V$

$$b) V_o = -\frac{R_f}{10k} V_i - \frac{R_f}{10k} V_i$$

$$\therefore \frac{V_o}{V_i} = -\frac{10k}{10k} - \frac{10k}{10k}$$

$$= \underline{\underline{-2}}$$

or at source V_i $10k \parallel 10k$
 $\therefore R_i = 5k\Omega$

$$V_o = -\frac{10k}{5k} V_i$$

$$\therefore \underline{\underline{\frac{V_o}{V_i} = -2}}$$

$$c) I_1 = I_o + I_f$$

$$\therefore I_o = I_1 - I_f$$

$$= \frac{V_o}{10k} - \frac{V_a - V_o}{10k}$$

$$\text{but } V_o = -2V_i = -20V$$

$$\text{Also } V_a = 0V$$

$$\therefore I_o = \frac{-20}{10k} - \frac{0 - (-20)}{10k}$$

$$= -2m - 2m$$

$$\underline{\underline{I_o = -4m}}$$

Question 2.3 [4]

- a) Determine V_0 when $R_1 = 10\text{k}\Omega$, $R_2 = 20\text{k}\Omega$, and $R_4 = 3\text{k}\Omega$ in Fig Q2.3.
 b) If $R_4 = 6\text{k}\Omega$, determine R_1 and R_2 that will result in $V_0 = 0.2\text{ V}$.

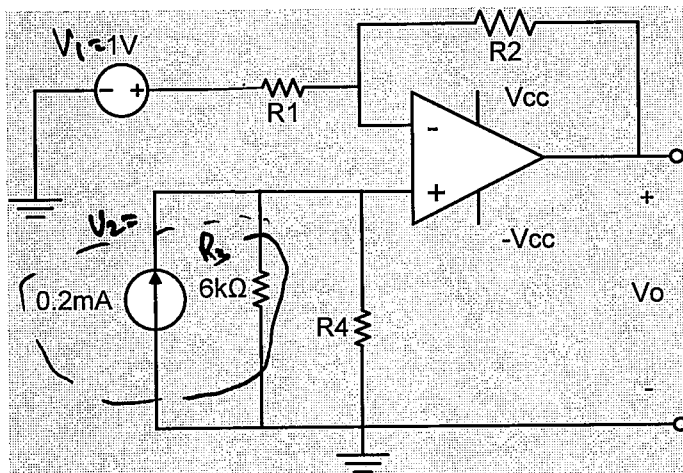
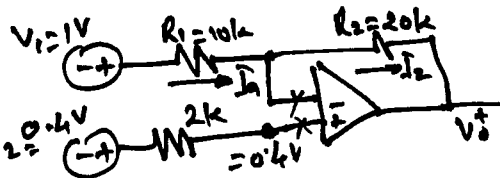


Fig. Q2.3

Another way of solving it:
 at Non-Inverting.

$$6\text{k} \parallel R_4 = \frac{6\text{k} \cdot 3\text{k}}{6\text{k} + 3\text{k}} = 2\text{k}\Omega$$

$$V_2 = 0.2\text{mA} \cdot 2\text{k} = 0.4\text{V}$$

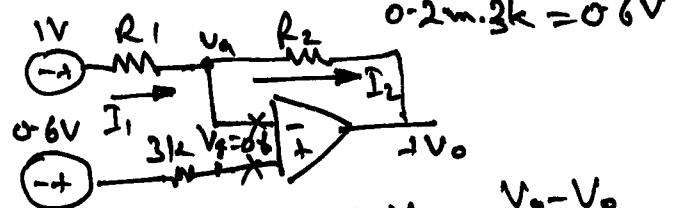
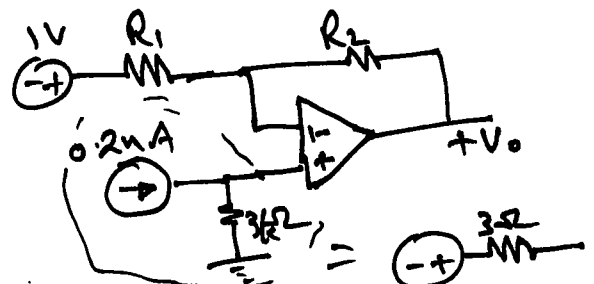


$$I_1 = I_2 \Rightarrow \frac{1 - 0.4}{10\text{k}} = \frac{0.4 - V_0}{20\text{k}}$$

$$1.2 = 0.4 - V_0 \Rightarrow \underline{\underline{V_0 = -0.8\text{V}}}$$

b) $R_3 = 6\text{k}$ and $R_4 = 6\text{k}$

$$\therefore R = \frac{6\text{k} \cdot 6\text{k}}{6\text{k} + 6\text{k}} = 3\text{k}\Omega$$



$$I_1 = I_2 \Rightarrow \frac{1 - V_a}{R_1} = \frac{V_a - V_0}{R_2}$$

$$\underline{\underline{V_a = 0.6}}$$

$$\therefore \frac{1 - 0.6}{R_1} = \frac{0.6 - 0.2}{R_2}$$

$$\frac{0.4}{R_1} = \frac{0.4}{R_2}$$

$$\therefore \underline{\underline{R_1 = R_2}}$$

Differential

$$a) \quad V_0 = \frac{R_2}{R_1} \left[\frac{1 + \frac{R_1}{R_2}}{1 + \frac{R_2}{R_4}} V_2 - V_1 \right]$$

$$V_1 = 1\text{V}, \quad V_2 = 0.2\text{mA} \cdot 6\text{k} = 1.2\text{V}$$

$$\therefore V_0 = \frac{20\text{k}}{10\text{k}} \left[\frac{1 + \frac{10\text{k}}{20\text{k}}}{1 + \frac{6\text{k}}{3\text{k}}} \cdot 1.2 - 1 \right]$$

$$= 2 \left[\frac{3/2}{5} \cdot 1.2 - 1 \right]$$

$$\underline{\underline{V_0 = -0.8\text{V}}}$$

$$b) \quad \frac{R_3}{R_4} = \frac{R_1}{R_2}, \quad \frac{R_3}{R_4} = \frac{6\text{k}}{6\text{k}} = 1$$

$$\therefore V_0 = \frac{R_2}{R_1} (V_2 - V_1)$$

$$0.2 = \frac{R_2}{R_1} (1.2 - 1)$$

$$0.2 = \frac{R_2}{R_1} (0.2)$$

$$\therefore \underline{\underline{R_1 = R_2}}$$

Question 2.4 [4]

Determine the following for Fig Q2.4:

- a) V_{o1} b) V_{o2} c) V_o

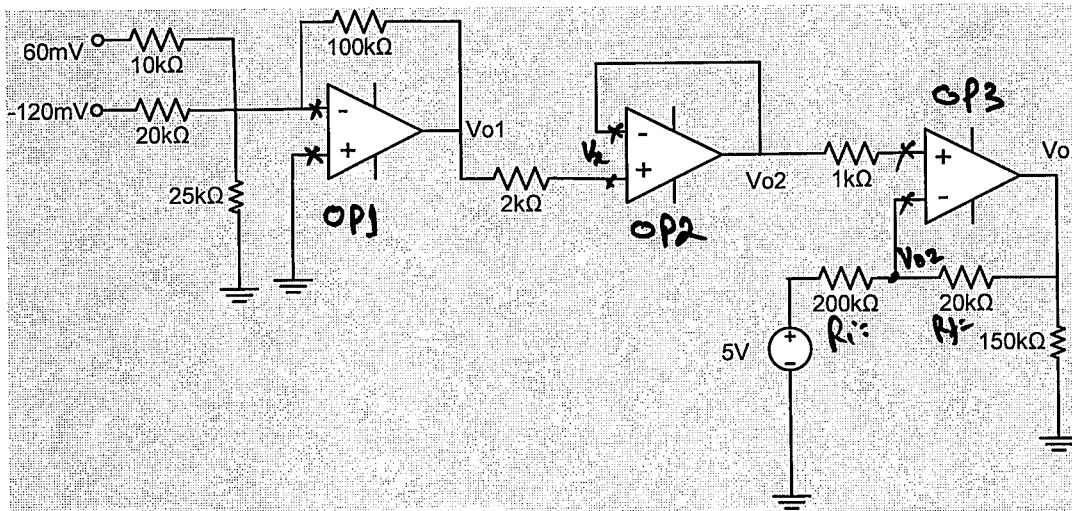


Fig. Q2.4

a) OP1: Summer:

$$\therefore V_{o1} = - \left[\frac{100k}{10k} \cdot 60m + \frac{100k}{20k} \cdot (-120m) + \frac{100k}{25k} \cdot 0 \right]$$

$$= - [600m + (-600m)]$$

$$\therefore \underline{\underline{V_{o1} = 0}}$$

b) OP2: Voltage Follower

$$\therefore V_{o2} = V_{ox}, \text{ but } V_{ox} = V_{o1}$$

$$\therefore \underline{\underline{V_{o2} = V_{o1} = 0V}}$$

c) OP3: Inverter

$$\therefore V_o = - \frac{R_f}{R_i} V_i$$

$$= - \frac{20k}{20k} \cdot 5$$

$$\underline{\underline{V_o = -0.5V}}$$

Question 3.1 [4]

Use linearity to determine the following for the circuit in Fig. Q3.1:

- a) V_s if $V_0 = 1\text{ V}$
- b) V_0 if $V_s = 20\text{ V}$

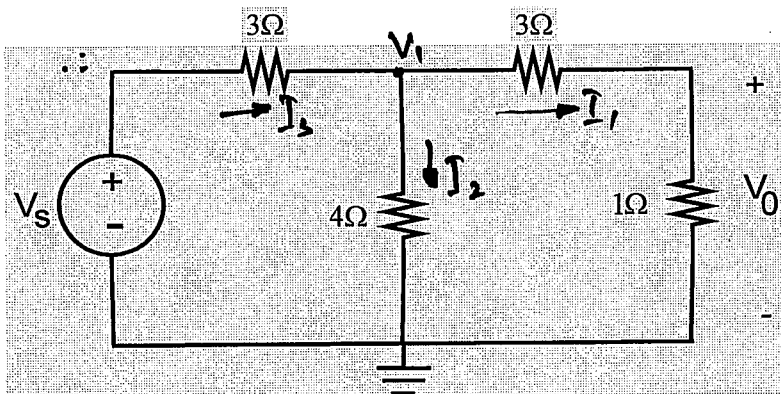


Fig. Q3.1

a) $V_0 = 1, \therefore I_1 = \frac{V_0}{1} = 1\text{ A}$

$\therefore V_1 = 1 \cdot (3 + 1) = 4\text{ V}$

$I_2 = \frac{4}{4} = 1\text{ A}$

$I_3 = 1 + 1 = 2\text{ A}$

$\therefore V_s = V_1 + 3 \cdot I_3$
 $= 4 + 3 \cdot 2$

$V_s = 10\text{ V}$

b) $V_s = 20$

$\therefore \text{Linearity factor} = \frac{20}{10} = 2$

$\therefore V_0 = 1 \times 2 = \underline{\underline{2\text{ V}}}$

Question 3.2 [4]

Use superposition to find the following for the circuit in Fig. Q3.2:

- V_0 due to the 3V source only
- V_0 due to the 9V source only

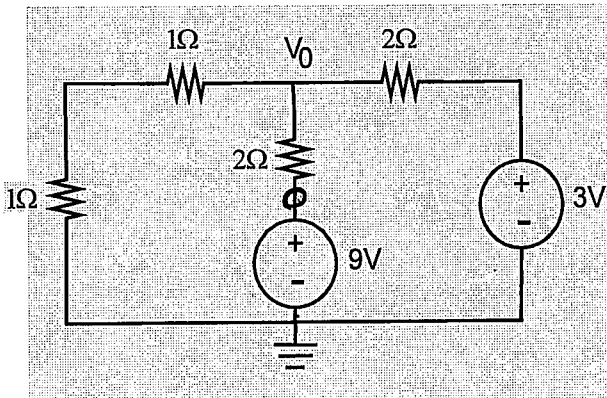
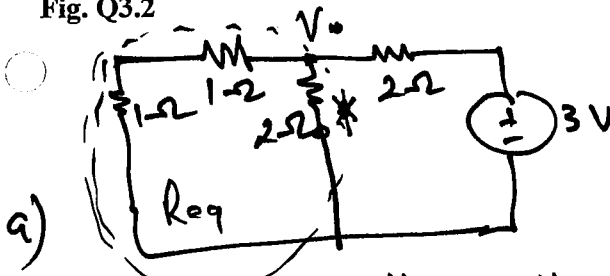


Fig. Q3.2



$$R_{eq} = (1+1) \parallel 2 = 2 \parallel 2$$

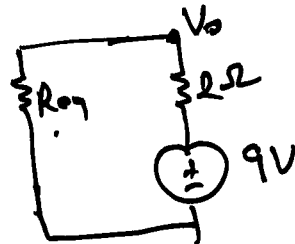
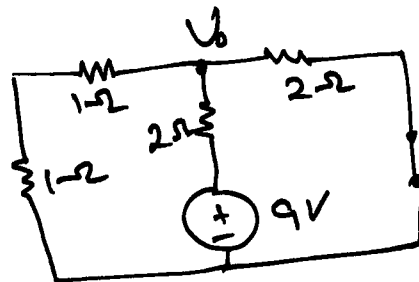
$$\therefore R_{eq} = 1\Omega$$

Using Voltage divider

$$V_0 = \frac{R_{eq}}{2 + R_{eq}} \cdot 3 = \frac{1}{2+1} \cdot 3$$

$$\therefore \underline{V_0 = 1V}$$

(b)



$$R_{eq} = (1+1) \parallel 2 = 2 \parallel 2 = 1\Omega$$

$$\therefore V_0 = \frac{R_{eq}}{R_{eq} + 2} \cdot 9 = \frac{1}{2+1} \cdot 9 = 3V$$

$$\therefore \underline{V_0 = 3V}$$

Question 3.3 [2]

Determine V_x for the circuit in **Fig. Q3.3** using source transformations.

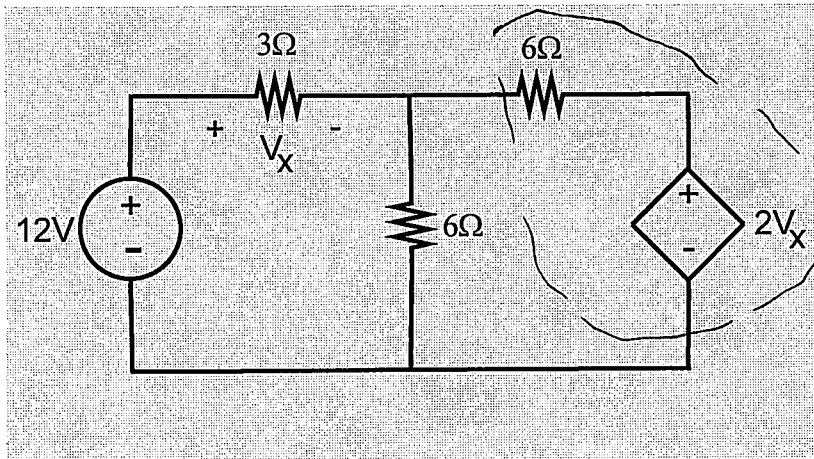
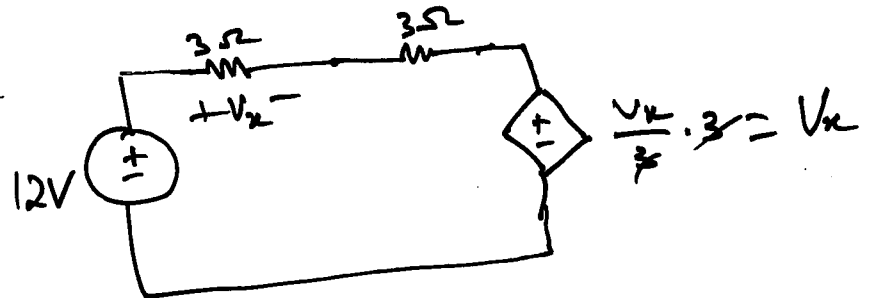
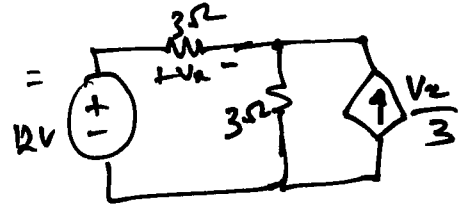
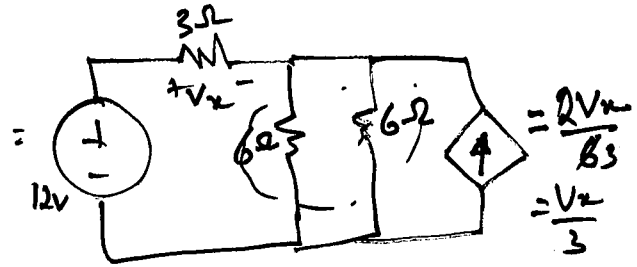


Fig. Q3.3



$$\therefore -12 + V_x + V_x + V_x = 0$$

$$\therefore \boxed{V_x = 4V}$$

Question 3.4 [3]

- a) Determine the Thevenin equivalent of the circuit in Fig. Q3.4.
b) What load resistance will result in maximum power transfer?

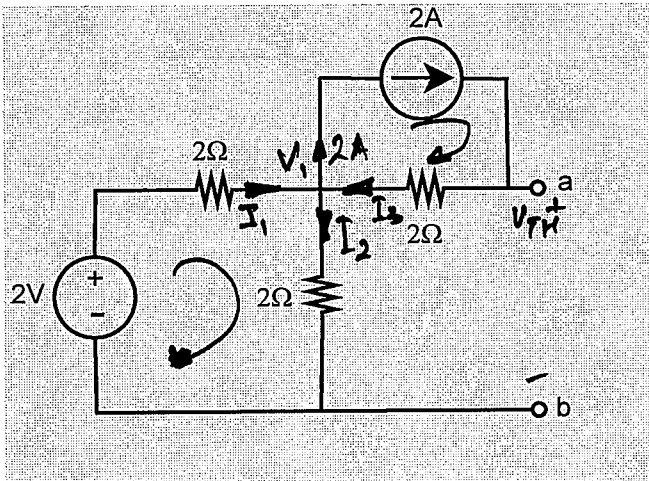
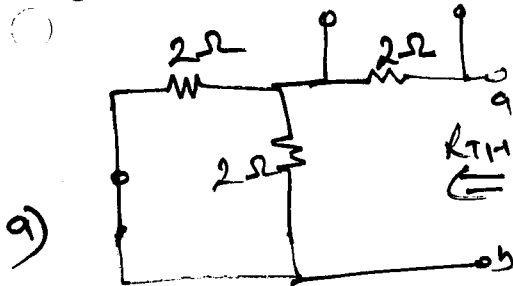
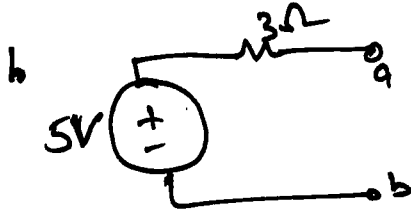


Fig. Q3.4



$$\therefore R_{TH} = 2 + (2 \parallel 2)$$

$$\underline{R_{TH} = 3\Omega}$$



b)

$$P_{max} = \frac{V_{TH}^2}{4 R_{TH}}$$

$$\underline{R_L = R_{TH} = 3\Omega}$$

e Node V_1 .

$$I + I_3 - 2 - I_2 = 0$$

$$\frac{2 - V_1}{2} + I_3 - 2 - \frac{V_1}{2} = 0, \text{ but } I_3 = 2A$$

$$\therefore \frac{2 - V_1}{2} - \frac{V_1}{2} = 0$$

$$\therefore V_1 = 1V$$

$$\therefore V_{TH} =$$

$$\text{Also } I_3 = \frac{V_{TH} - V_1}{2}$$

$$2 = \frac{V_{TH} - 1}{2}$$

$$\therefore V_{TH} = 4 + 1 = \underline{\underline{5V}}$$

Question 3.5 [2]

Determine the Norton equivalent resistance wrt terminals a-b for the circuit in Fig. Q3.5.

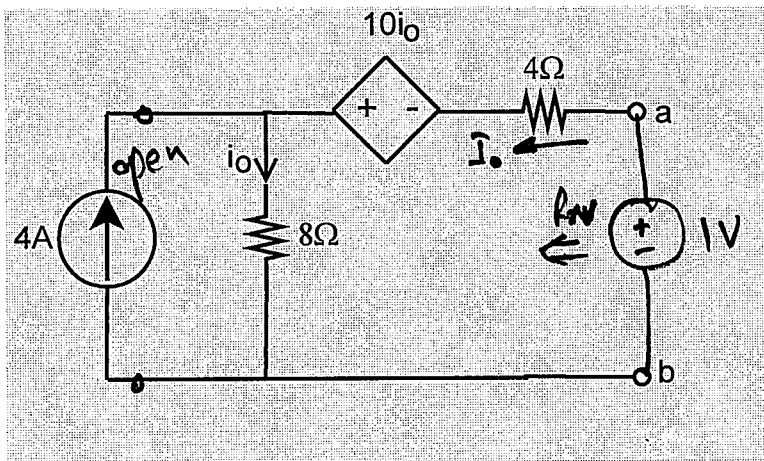


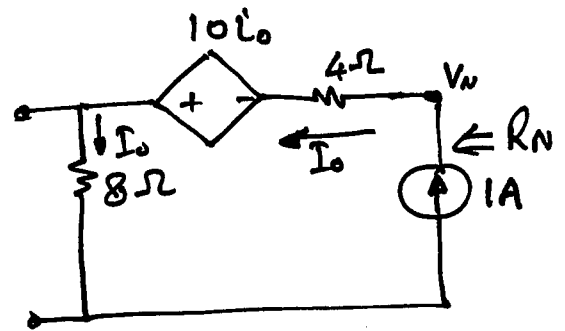
Fig. Q3.5

$$-1 + 4I_o - 10I_o + 8I_o = 0$$

$$2I_o = 1$$

$$\therefore I_o = 1/2 \text{ A}$$

$$\therefore R_N = \frac{1}{I_o} = \frac{1}{1/2} = \underline{\underline{2 \Omega}}$$



$$I_o = 1 \text{ A}$$

$$-V_N + 4I_o - 10I_o + 8I_o = 0$$

$$\therefore V_N = 12 - 10 = 2 \text{ V}$$

$$\therefore R_N = \frac{V_N}{1} = \underline{\underline{2 \Omega}}$$